Fakultas Informatika
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Aplikasi Berbasis Platform

HTML

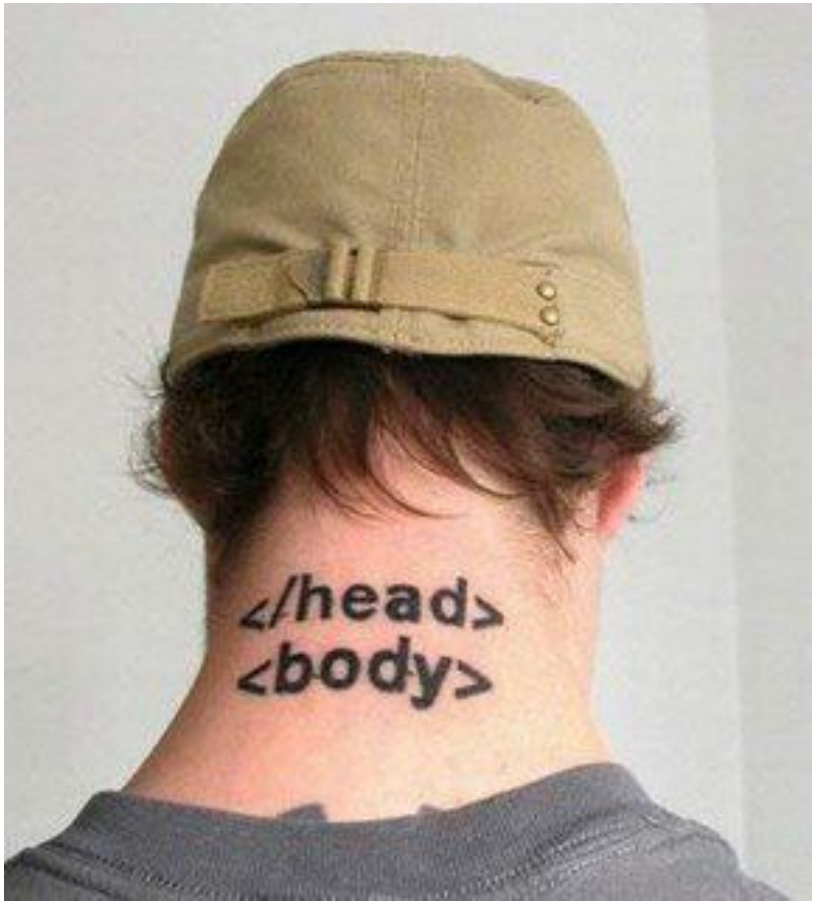
HTML

- Describe content and defines the **structure** within webpage
- Hypertext documents on the web are designed to be linked to other documents
- Markup language, using specific text sequence (tag) to interpret specific visualization
- HTML file is a text file, not binary file, so you can open and edit it with any text editor
- Doesn’t need to be compiled

Basic HTML Syntax

```
<!DOCTYPE html>
<html>
    <head>
            <title>Page title</title>
    </head>
    <body>
            Hello World
    </body>
</html>
```

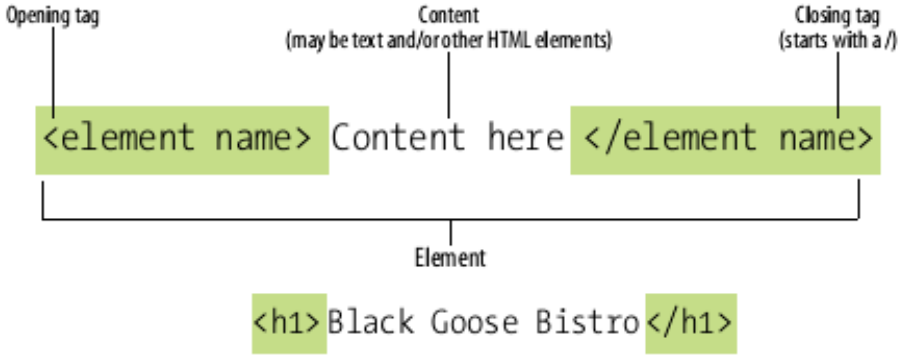
- HTML tag: mark the starting and ending point for HTML document
- HEAD tag: header part of HTML document, usually contains page title and external linked file (CSS file for styling, JS file for functions)
- TITLE tag: page title for this HTML document, will show up at topleft corner of browser
- BODY tag: content part of HTML document. all tags for visualization is put inside here


</head>
<body>

Basic HTML element (tag) structure

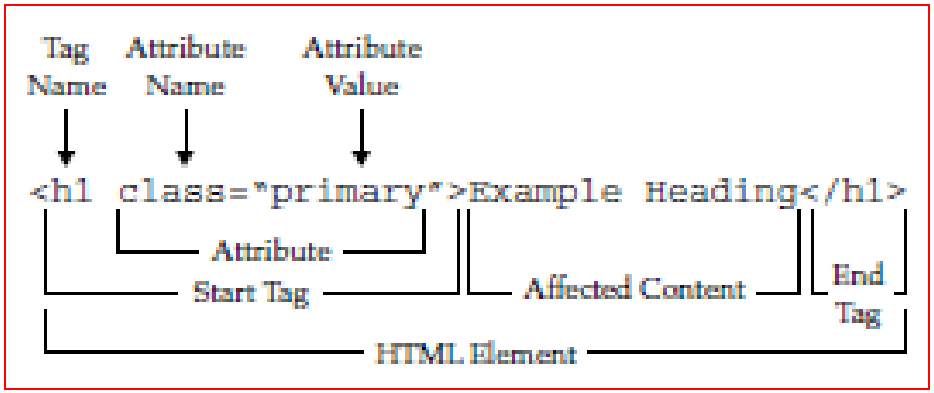
- Mostly HTML tags have opening and closing tag, but some tags aren't
  - <img />
  -
- Tag writing must be ordered like parenthesis, don't overlap each other

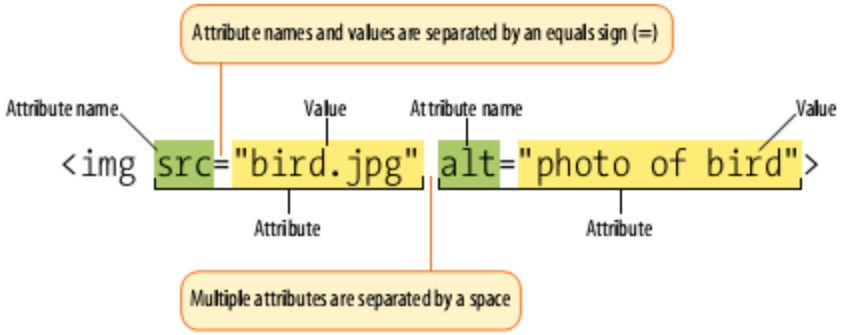
```
<p>
    <b>
        …
    </b>
</p>
```



Basic HTML element (tag) structure

- Attribute : define the property of HTML element





Basic HTML tags

- Line break :
- Paragraph : ....
- Bold : ... , Italic : ... , Underline : <u> ... </u>
- Heading : <h1> ... </h1> → <h6> ... </h6>
- Horizontal line : <hr width="100%" noshade />
- Attach image : <img src="…" />
- Link : <a href="…"> … </a>
- Bullets List: <ul> … </ul>
- Numbering List : <ol> … </ol>
- Ignored tag (usually for code commenting) : <!-- …. -->

HTML Tags Cheatsheet

Page sections	
address	author contact information
article (5)	self-contained content
aside (5)	tangential content (sidebar)
footer (5)	related content
header (5)	introductory content
nav (5)	primary navigation
section (5)	conceptually related group of content

Heading content	
h1...h6	headings, levels 1 through 6
hgroup	heading group

Grouping content	
blockquote	blockquote
div	generic division
figure (5)	related image or resource
figcaption (5)	text description of a figure
hr	paragraph-level thematic break (horizontal rule)
p	paragraph
pre	preformatted text

List elements	
dd	definition
dl	definition list
dt	term
li	list item (for ul and ol)
ol	ordered list
ul	unordered list

Breaks	
br	line break
wbr (5)	word break

Element	Description
a	An anchor or hypertext link (see Chapter 6 for details)
abbr	Abbreviation
b	Added visual attention, such as keywords (bold)
bdi	NEW IN HTML5 Indicates text that may have directional requirements
bdo	Bidirectional override; explicitly indicates text direction (left to right, ltr, or right to left, rtl)
br	Line break
cite	Citation; a reference to the title of a work, such as a book title
code	Computer code sample
data	WHATWG ONLY Machine-readable equivalent dates, time, weights, and other measurable values
del	Deleted text; indicates an edit made to a document
dfn	The defining instance or first occurrence of a term
em	Emphasized text
i	Alternative voice (italic)
ins	Inserted text; indicates an insertion in a document
kbd	Keyboard; text entered by a user (for technical documents)
mark	NEW IN HTML5 Contextually relevant text
q	Short, inline quotation
ruby, rt, rp	NEW IN HTML5 Provides annotations or pronunciation guides under East Asian typography and ideographs
s	Incorrect text (strike-through)
samp	Sample output from programs
small	Small print, such as a copyright or legal notice (displayed in a smaller type size)
span	Generic phrase content
strong	Content of strong importance
sub	Subscript
sup	Superscript
time	NEW IN HTML5 Machine-readable time data
u	Underlined
var	A variable or program argument (for technical documents)
wbr	Word break

Heading

```
<h1>Title 1</h1>

<h2>Title 2</h2>

<h3>Title 3</h3>

<h4>Title 4</h4>

<h5>Title 5</h5>

<h6>Title 6</h6>
```

Title 1

Title 2

Title 3

Title 4

Title 5

Title 6

Paragraph

```
<p>
  This is my first page
</p>

<p>This page is under
development</p><p>I will try
hard to improve this page</p>
```

This is my first page

This page is under development

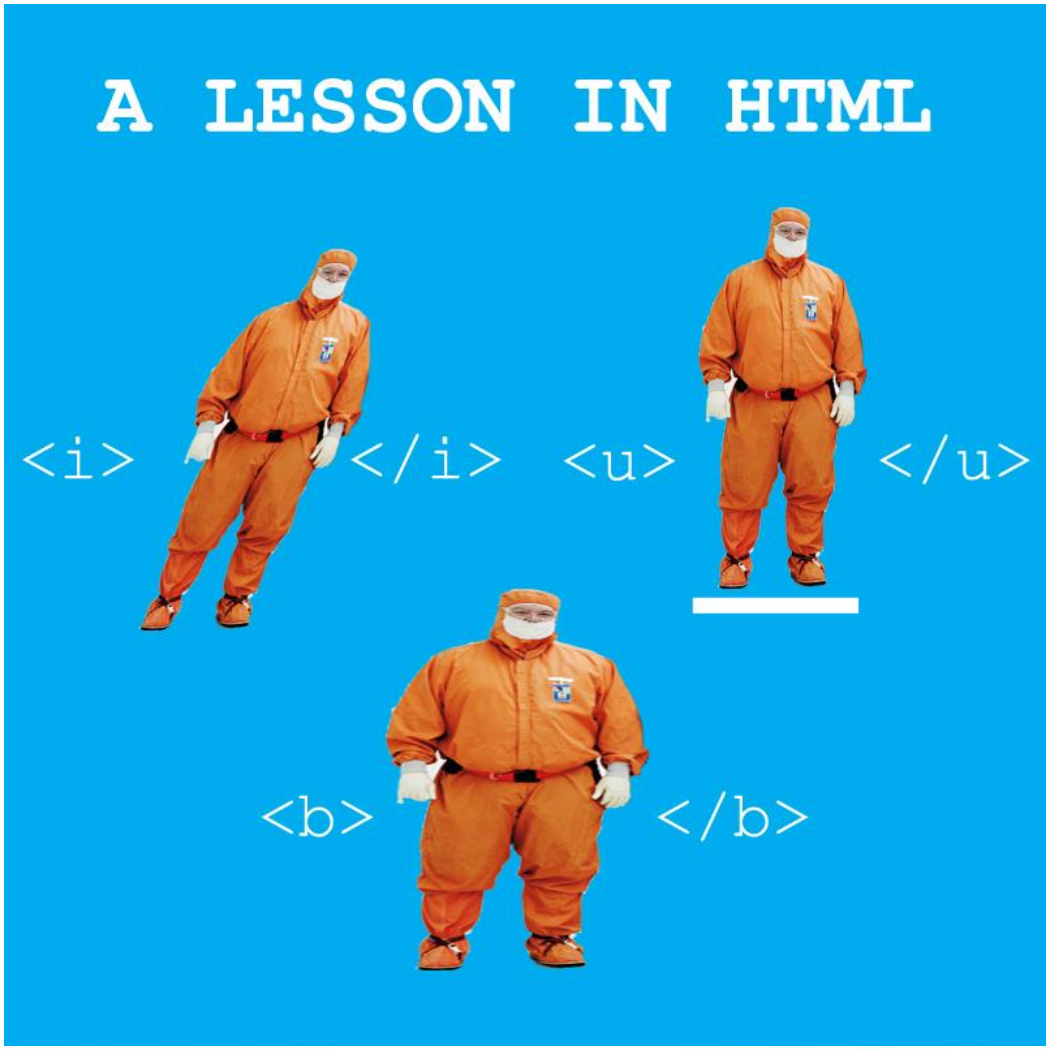
I will try hard to improve this page

<h1>he doc</h1>
 how are u

Font Styling

```
<p>You can print <b>bold</b>,
<i>italic</i>, <u>underline</u>,
<b>bold then <i>italic and
bold</i></b>, or
<b><i><u>combination</u></i></b> styles
between normal styles</p>
```

You can print **bold,** *italic*, <u>underline</u>, **bold then *italic and bold***, or ***<u>combination</u>*** styles between normal styles


A LESSON IN HTML

<u>
</u>

Line break

```
<p>Hello,<br />
my name is … <br />Welcome to my homepage</p>

<p>This page <br /> is under development</p>
```

Hello,
my name is .....
Welcome to my homepage

This page
is under development

List

- Ordered list / Numbering (A/a=alphabet, I/i=roman, 1=number)

```
<ol type="A|a|I|i|1">
   <li> first </li>
   <li> second </li>
   <li> .... </li>
</ol>
```

- Unordered list / Bullets

```
<ul type="circle|square|disc">
   <li> .... </li>
</ul>
```

List

```
Expert group:
<ul>
   <li>SE</li>
   <li>IS</li>
   <li>CPS</li>
</ul>

Study Programs:
<ol>
  <li>Master of Informatics</li>
      Graduate program of Informatics
Engineering
  <li>Bachelor</li>
  <ul>
      <li>Software Engineering</li>
      <li>Information Technology</li>
  </ul>
</ol>
```

Expert group:

- SE
- IS
- CPS

Study Programs:

1. Master of Informatics
   Graduate program of Informatics Engineering
2. Bachelor
   - o Software Engineering
   - o Information Technology

Horizontal line

```
Show the line!<hr />
Blahblah
```

Show the line!

Blahblah

Attach Image

```
<img src="cover.jpg" align="left" />
```
Add attribute 'align' to position the image. Add attribute 'alt' for showing text if image is unloaded.
```
<img src="logo.gif" alt="Logo" width="100" height="100" />
```
Add attribute 'width' and 'height' for stretching image to desired size.

Add attribute align to position the image. Add attribute alt for showing text if image is unloaded. Add attribute 'width' and 'height' for streching image to desired size.

Hyperlink

- Link to your other page:

```
Next: <a href="page2.html">Page 2</a>
```

- Link to another site:

```
klik link ini <a href="http://google.com">google</a>
```

klik link ini google

- Link with image:

klik link pada gambar ini

```
klik link pada gambar ini

<a href="http://google.com"><img src="img.png"
alt="text image" /></a>
```

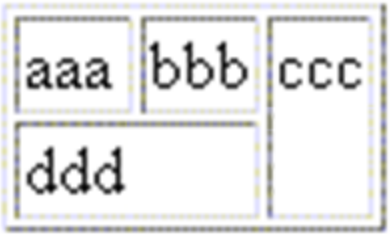
Table

```
<table border="1">
  <tr> <!-- row 1 -->
       <th>aaa</th> <!-- col 1 at row 1 -->
       <th>bbb</th> <!-- col 2 at row 1 -->
       <th>ccc</th> <!-- col 3 at row 1 -->
  </tr>
  <tr>
       <td>ddd</td>
       <td>eee</td>
       <td>fff</td>
  </tr>
</table>
```

aaa	bbb	ccc
ddd	eee	fff

Table with merged columns / rows

```
<table border="1">
  <tr>
     <th>aaa</th>
     <th>bbb</th>
     <th rowspan="2">ccc</th>
  </tr>
  <tr>
     <td colspan="2">ddd</td>
  </tr>
</table>
```



Exercise

Nama Lengkap	Gelar Pendidikan		Age
	Sarjana	Magister	
Budi	S.Kom	M.Sc	35
Andi	S.SiKom	M.T	52

Form

- Used for **handling input from user**
- Analogy: HTML document is a **fill out form**
- Usually, a form represents a specific task. Example: login, register, etc
- **A Form tag** can contain many tags for input with various data characteristics

```
<form method="..." action="...">
    ....
</form>
```

- Every **form** tag must specify the 'method' and 'action' attribute, to handle the data input sent (when you *submit* it). In every tag for input, 'name' attribute must be written to map that data when processed. We will discuss this later in Server Side Scripting

Example: Form

```
<form>
  First name:<br />
  <input type="text" name="firstname" /><br />
  Last name:<br />
  <input type="text" name="lastname" value="Mickey"><br /><br />
  <input type="submit" value="Submit" />
</form>
```

First name:

Last name:

Mickey

Submit

Tags inside Form

- single-line text field: <input type="text" />
- multi-line text field: <textarea></textarea>
- masked text field: <input type="password" />
- single selection:
  - datalist (text field with autocomplete)
  - radio button: <input type="radio" name="any" /> Yes
    <input type="radio" name="any" /> No
  - dropdown (combo box):

```
<select name="grade">
    <option value="A">Very good</option>
    <option value="AB">Good</option>
</select>
```

Tags inside Form

- multiple selection:

```
<select multiple name="grade">
        <option value="A">Very good</option>
        <option value="AB">Good</option>
</select>
```

- Check box: `<input type="checkbox" name="data[]" />` Coder

  `<input type="checkbox" name="data[]" />` Designer

- File Attach: `<input type="file" />`

- Buttons:

  Submit the form: `<input type="submit"  value="Send"/>`

  Reset the form: `<input type="reset" value="Reset" />`

  General button: `<input type="button" />`

  Another general button: `<button></button>`

Tags inside Form

- In <input> tag, there are many types in 'type' attribute

- **color**
- **date**
- **datetime**
- **datetime-local**
- **email**
- **month**
- **number**
- **range**
- **search**
- **tel**
- **time**
- **url**
- **week**

Example: Radio Button

```
<input type="radio" name="gender" value="male" checked /> Male <br />
<input type="radio" name="gender" value="female" /> Female
```

Male
Female

Example: Textarea

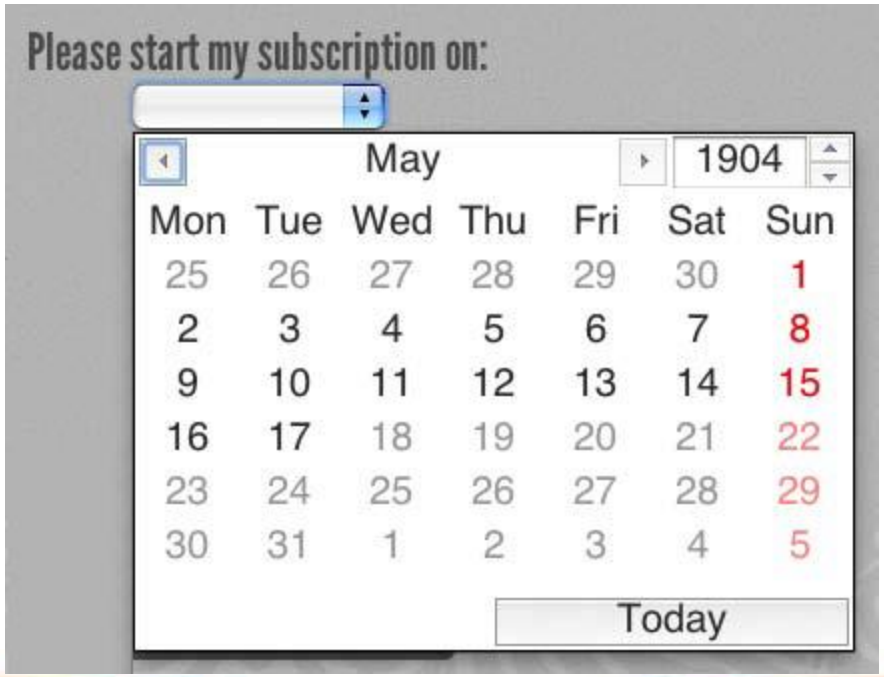
```
<textarea name="message" rows="10" cols="30">
The cat was playing in the garden.
</textarea>
```

The cat was playing in the garden.

Example: Date

```
Please start my subscription on:<br />

<input type="date" name="bday" />
```



Example: Datalist

```
Favorite Color:

<input type="text" list="colors" id="favcolor"
name="favcolor" />

<datalist id="colors">

  <option value="Blue">

  <option value="Green">

  <option value="Pink">

  <option value="Purple">

</datalist>
```

Example: Number

Quantity (between 1 and 5):

```
I would like to receive <input type="number" name="quantity" min="1" max="5" /> copies of The HTML5 Herald
```

Any question?

Always remember..

PROGRAMS ARE LIKE WOMEN,
NO MATTER HOW MUCH YOU GET MAD BECAUSE OF ERRORS,
YOU ARE ALWAYS THE ONE WHO IS WRONG

Exercise

Data Mahasiswa

NIM :

Nama :

Tgl Lahir (dd-mm-yyyy) : dd/mm/yyyy

Jenis Kelamin : Laki-Laki Perempuan

Alamat di Bandung :

Tinggal di : --pilih--
--pilih--
Kos/kontrakan
Rumah saudara
Rumah orang tua

Pilih 2 hobi yang paling disukai :
Science
Technology
Art
Sport

Submit Reset

References

- Lynda.com : Web Development Foundations: Full-Stack vs Front-End
- Stanford : http://web.stanford.edu/class/cs142/lectures.html
- https://www.w3schools.com/html/default.asp

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THANK YOU